non.unaids.remainder, step 2 (2026-11-02)

chain	seed	iterations.saved	acceptance.rate	final.log.likelihood	best.log.likelihood	errored.proposals
1	20263016	5000	0.277	−44935.6	−44907.6	0
2	20263017	5000	0.276	−44925.1	−44910.0	0
3	20263018	5000	0.275	−44940.9	−44917.2	0
4	20263019	5000	0.275	−44943.0	−44922.7	0

file
mcmc_non.unaids.remainder_step2_chain1_2026-10-30.Rdata
mcmc_non.unaids.remainder_step2_chain2_2026-10-30.Rdata
mcmc_non.unaids.remainder_step2_chain3_2026-10-30.Rdata
mcmc_non.unaids.remainder_step2_chain4_2026-10-30.Rdata

Worst R-hats (> 1.1 suggests the chains haven't converged):

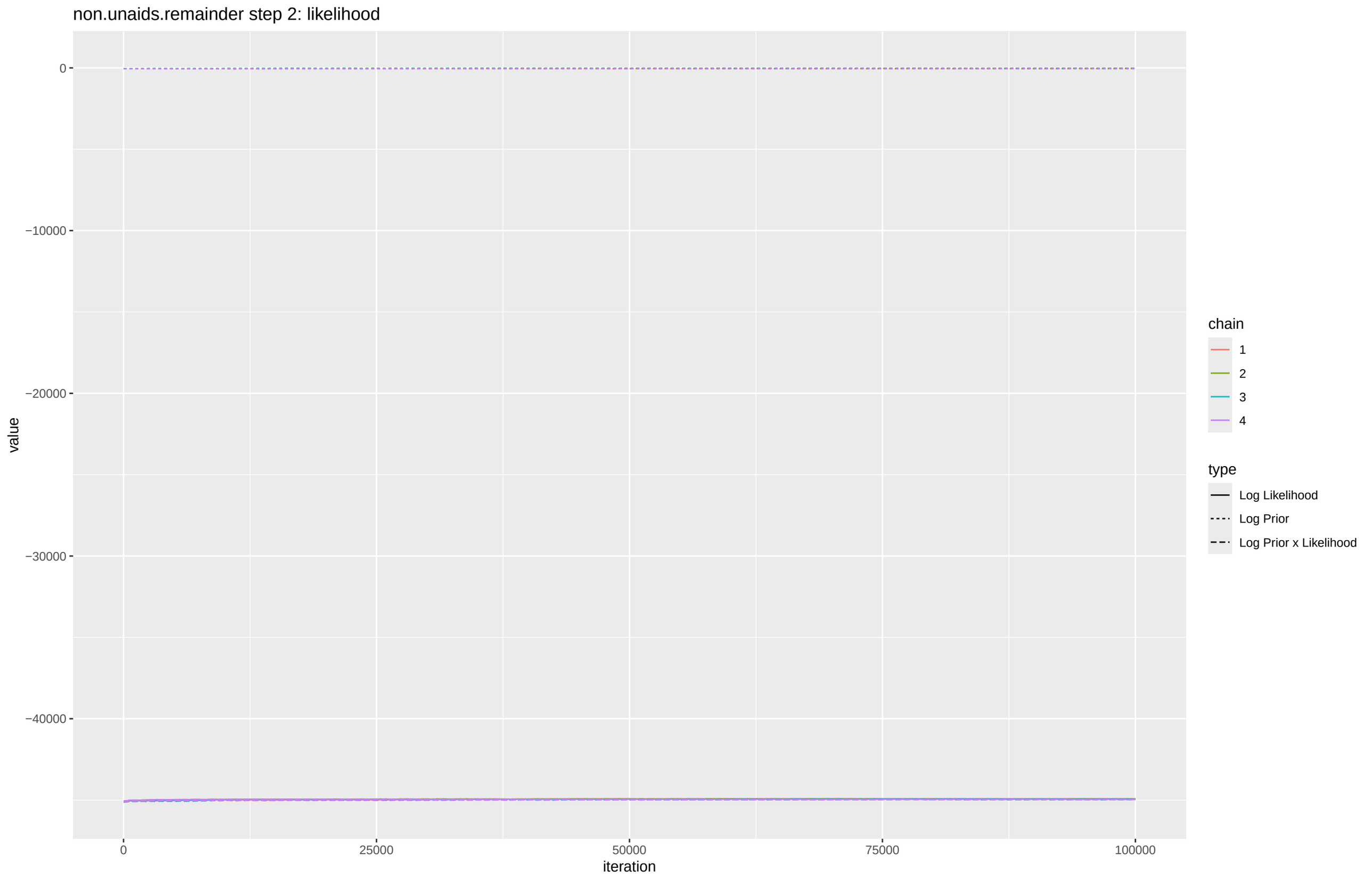
age.15.to.19.transmission.multiplier.0	2.416	age.20.to.29.transmission.multiplier.0	2.001
male.hiv.mortality.multiplier.1	1.723	age.65.to.79.mortality.slope.multiplier.male	1.661
female.to.male.multiplier	1.615	age.30.to.39.transmission.multiplier.3	1.566
male.awareness.multiplier	1.502	age.25.to.49.hiv.mortality.multiplier.0	1.493
male.hiv.mortality.multiplier.0	1.431	suppressed.disengagement.rates	1.427
age.15.to.24.hiv.mortality.multiplier.1	1.415	age.65.to.79.mortality.intercept.multiplier.male	1.389
trate.1	1.388	age.15.to.19.transmission.multiplier.1	1.382
age.20.to.29.transmission.multiplier.1	1.372	age.65.to.79.mortality.slope.multiplier.female	1.351
age.25.to.49.hiv.mortality.multiplier.2	1.341	age.25.to.49.hiv.mortality.multiplier.1	1.302
age.0.to.14.hiv.mortality.multiplier.2	1.302	trate.3	1.300
hiv.specific.mortality.rates.2	1.273	age.30.to.39.transmission.multiplier.0	1.270
trate.4	1.264	age.30.to.39.transmission.multiplier.1	1.258
age.30.to.39.transmission.multiplier.2	1.240		

Merged simset:

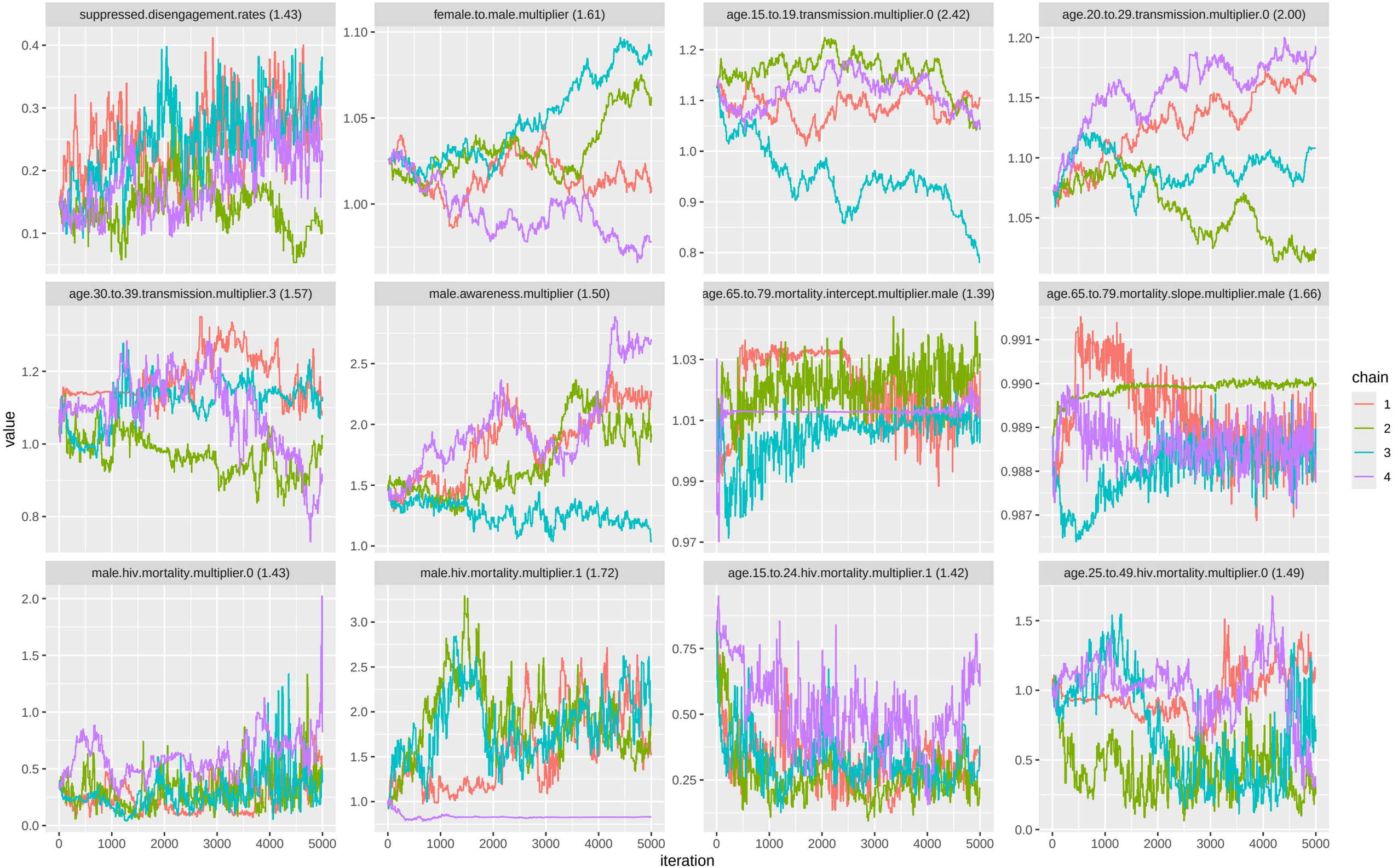
	location	simulations	total.weight	file
non.unaids.remainder	non.unaids.remainder	1000	1000	simset_non.unaids.remainder_step2_chains1234_2026-10-30.Rdata

Fit check (merged simset vs every calibration target, 2000–2025): 21 of 33 targets flagged

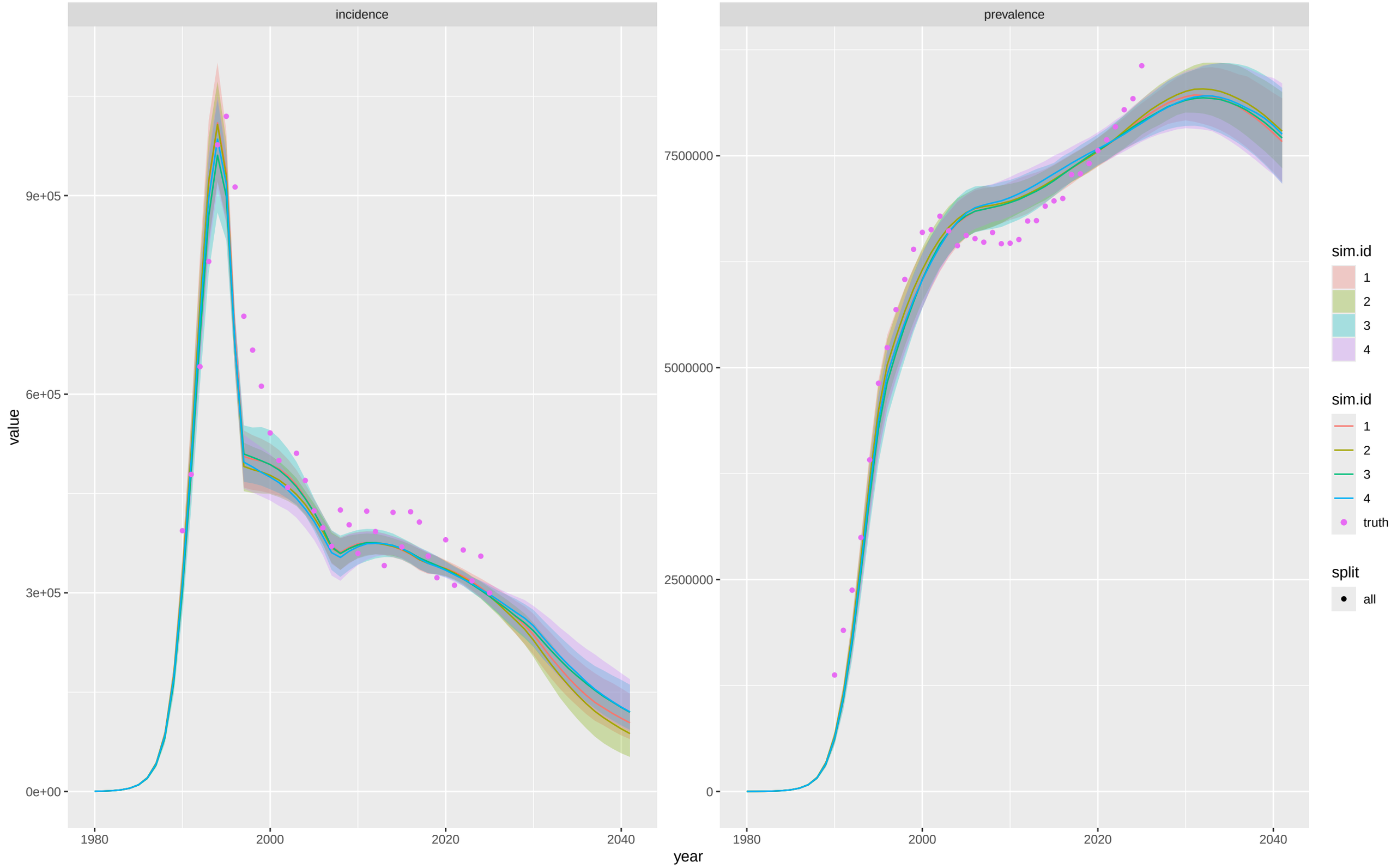
data.type	age	sex	years.with.data	pct.years.in.sim.95	median.sim.vs.data	last.year	last.sim.vs.data
hiv.mortality	15–24	both	20	0	x2.78	2019	x3.57
hiv.mortality	10–19	both	18	0	x2.14	2018	x4.05
hiv.mortality	50 and over	both	26	12	x1.71	2025	x2.18
prevalence	0–14	both	26	15	x0.78	2025	x0.65
hiv.mortality	All ages	both	26	4	x1.24	2025	x1.31
engagement	0–14	both	11	9	−10.0 pts	2025	−2.6 pts
hiv.mortality	25–49	both	20	20	x0.79	2019	x0.86
awareness	15+	female	11	0	−7.5 pts	2025	−7.5 pts
incidence	10–19	both	26	35	x0.70	2025	x0.70


non.unaids.remainder step 2: likelihood
0
−10000
−20000
−30000
−40000
lue
0
25000
50000
75000
100000
iteration
chain
1
2
3
4
type
Log Likelihood
Log Prior
Log Prior x Likelihood

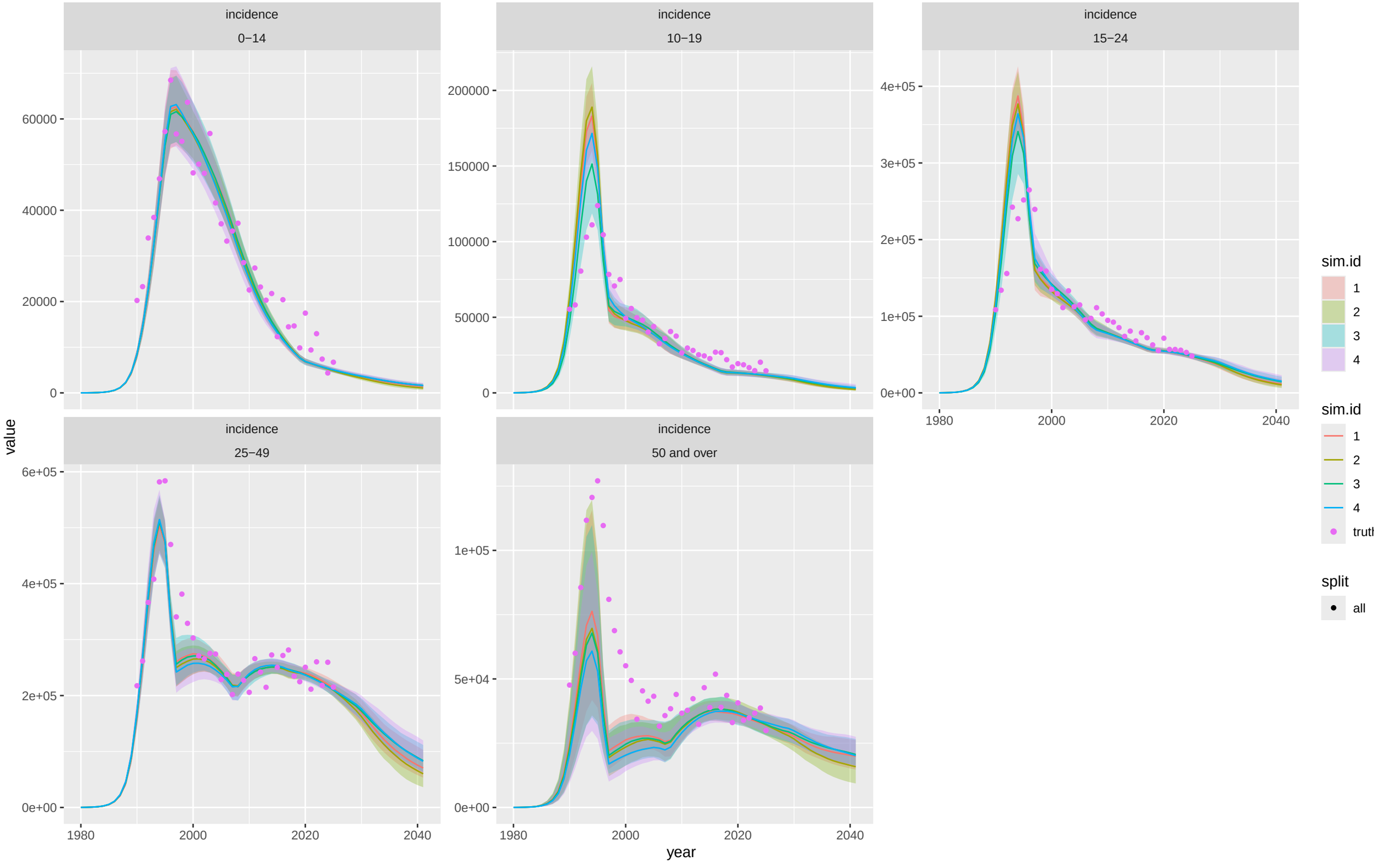
non.unaids.remainder step 2: trace



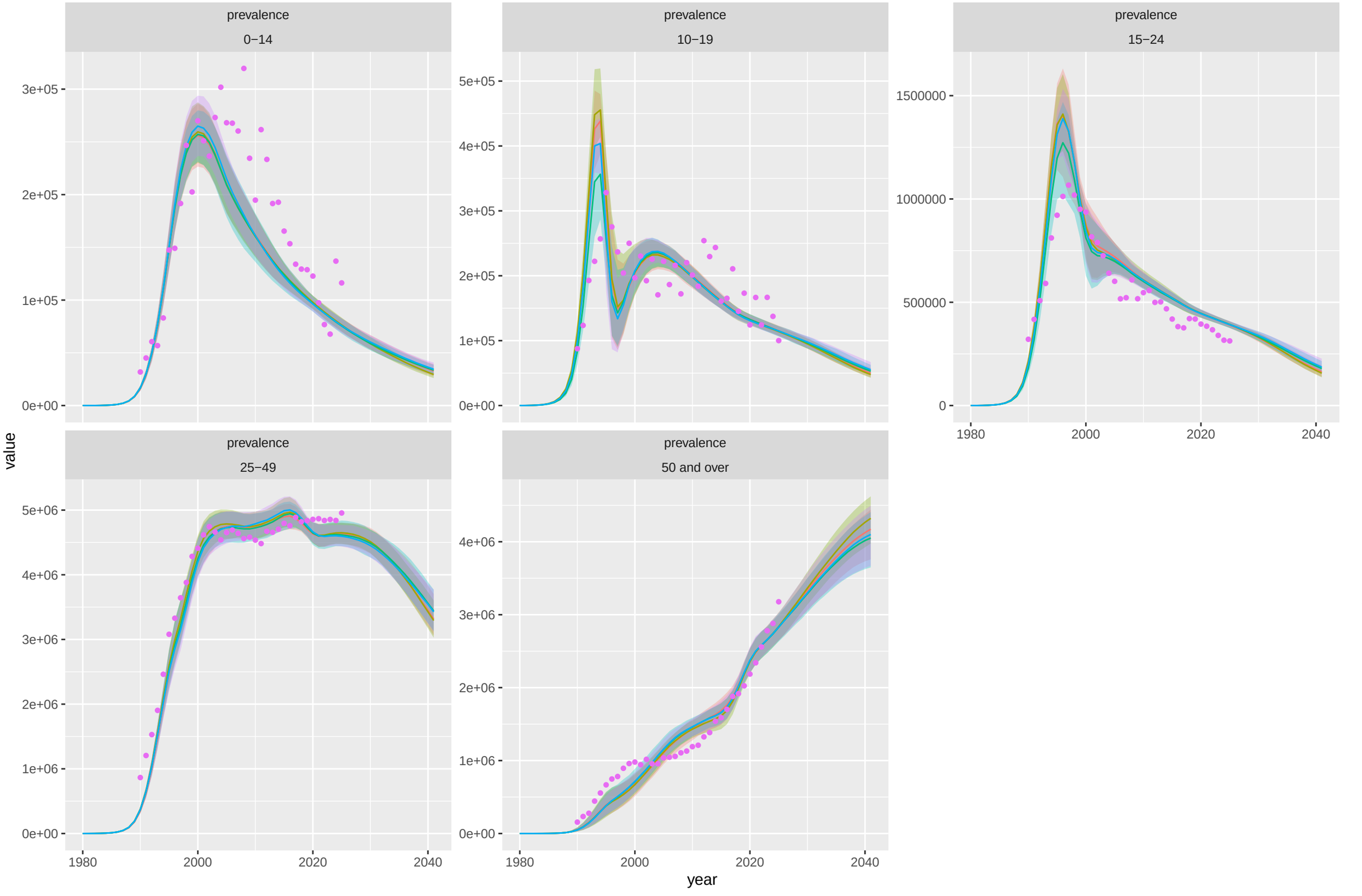
non.unaids.remainder step 2: fit (one color per chain)



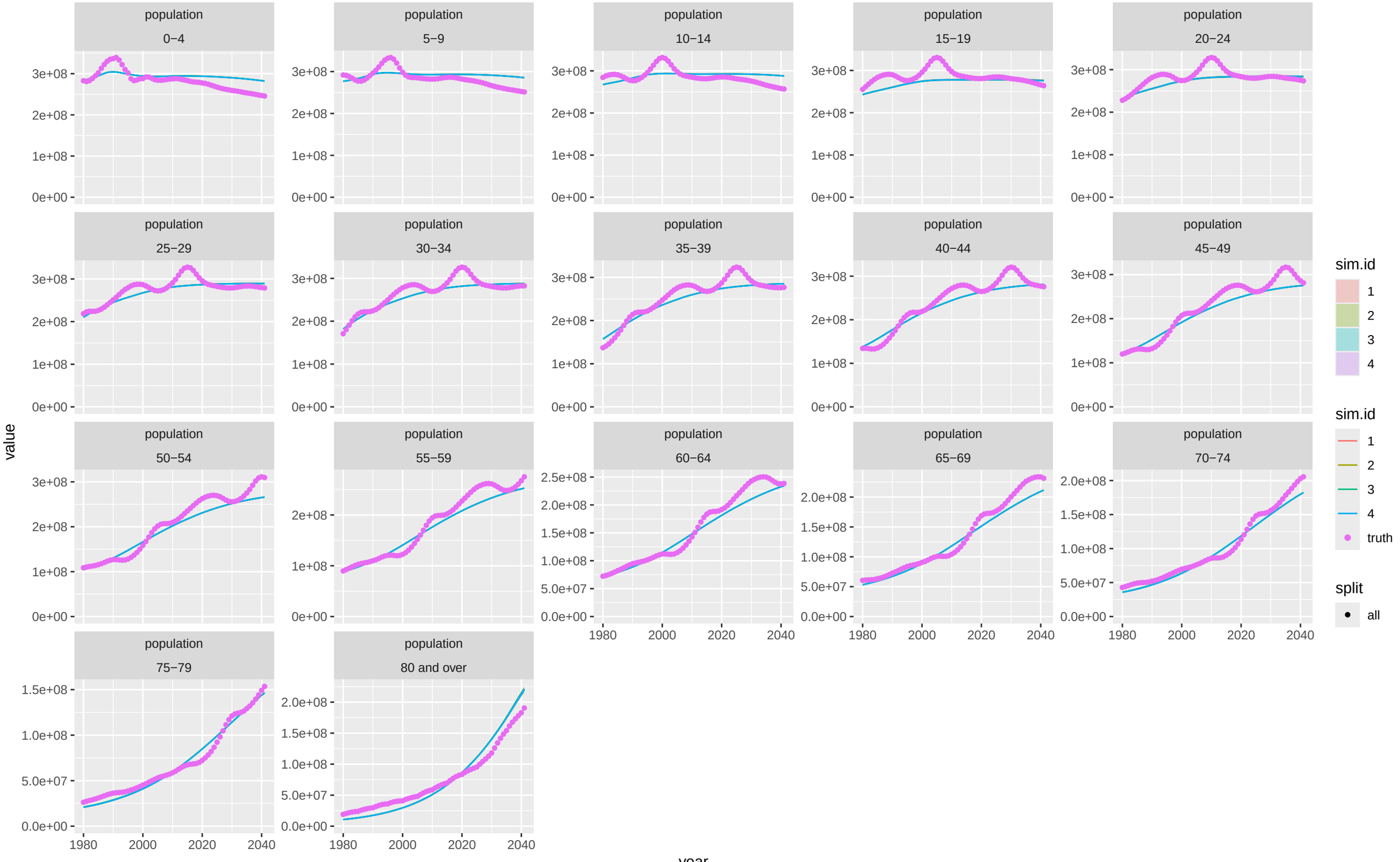
non.unaids.remainder step 2: incidence.by.age (one color per chain)



non.unaids.remainder step 2: prevalence.by.age (one color per chain)



non.unaids.remainder step 2: population.by.age (one color per chain)



non.unaids.remainder step 2: cascade (one color per chain)

awareness

engagement

suppression

value

year

1.00
0.75
0.50
0.25
0.00

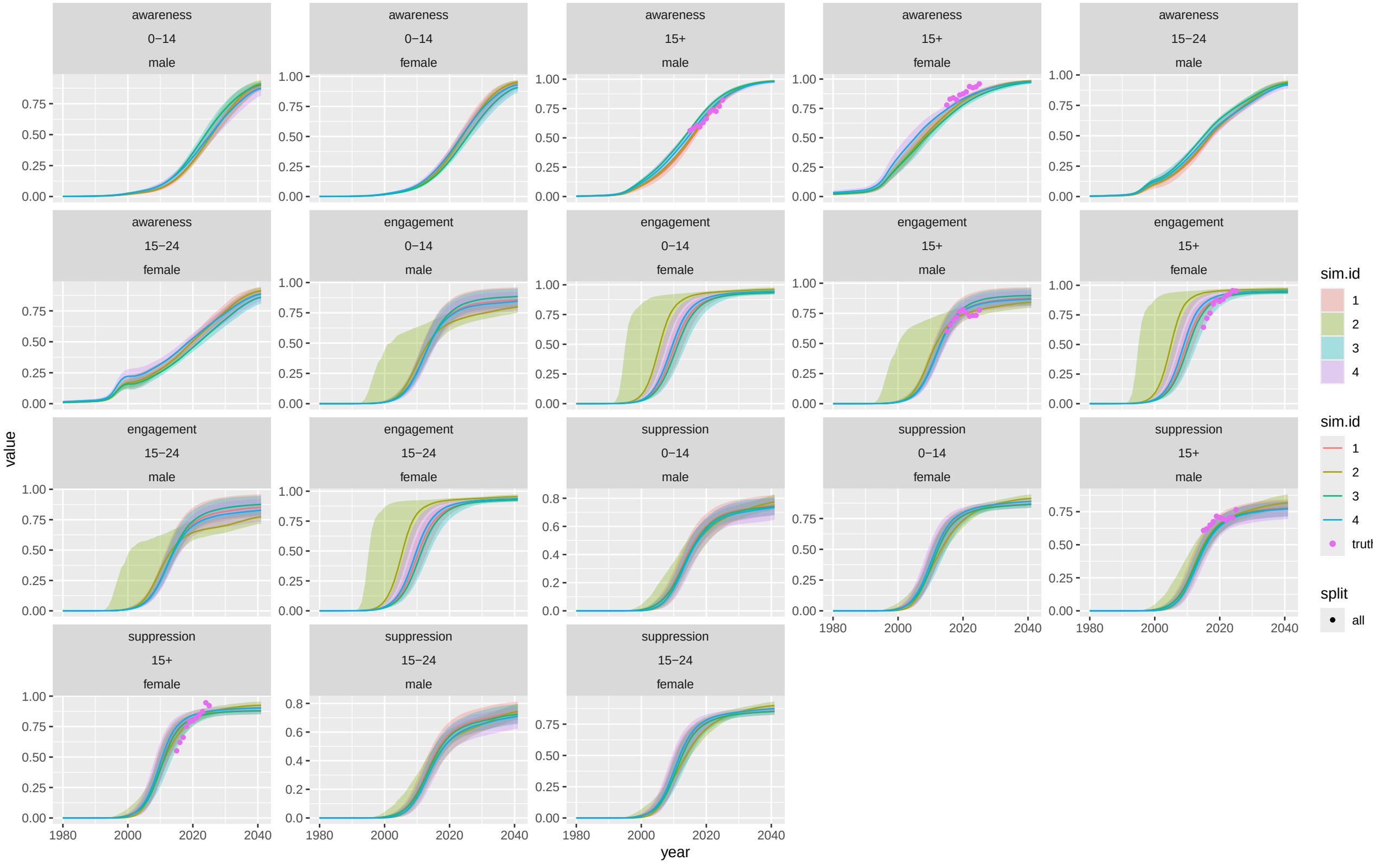
1980
2000
2020
2040

sim.id
1
2
3
4

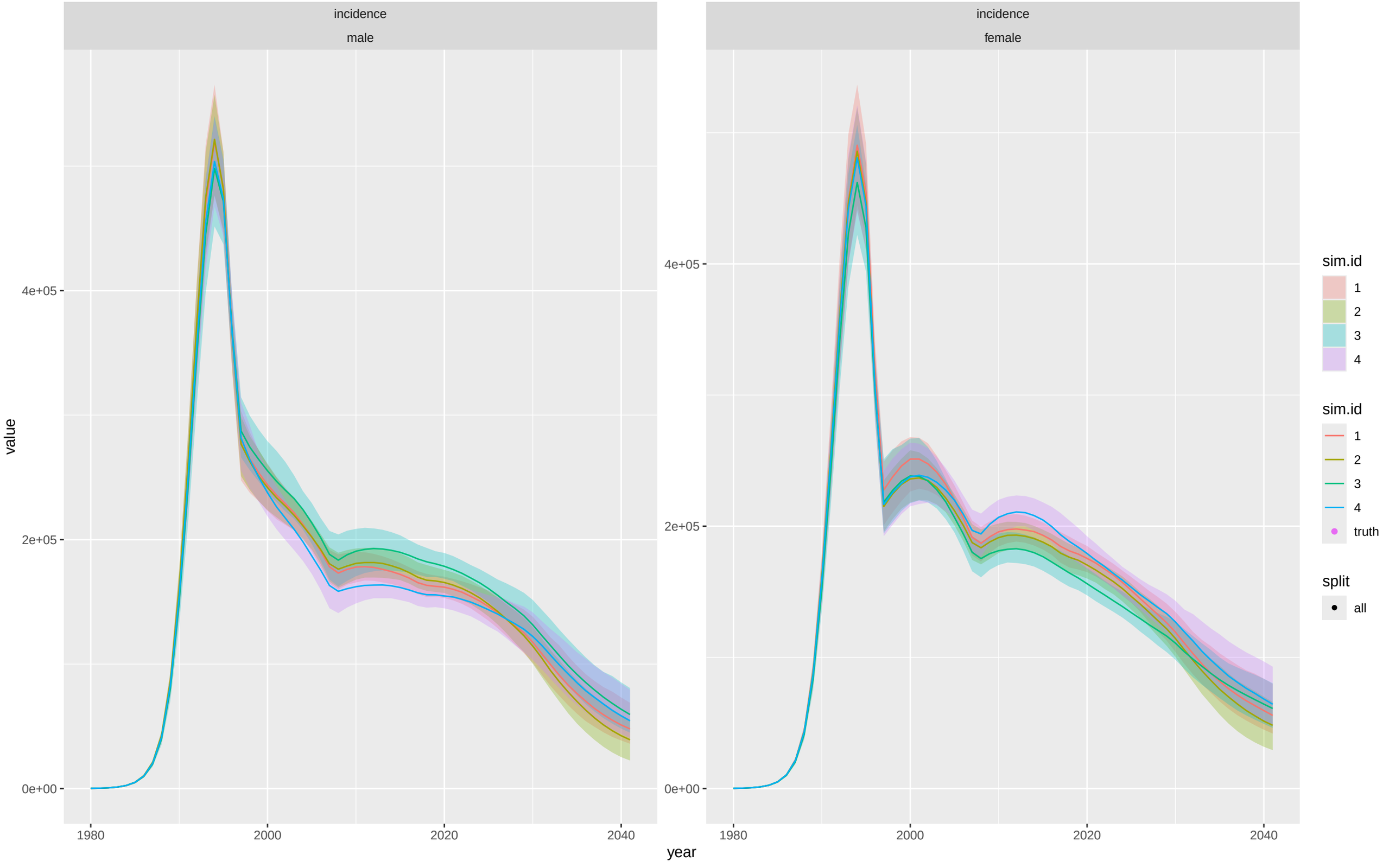
sim.id
1
2
3
4
truth

split
all

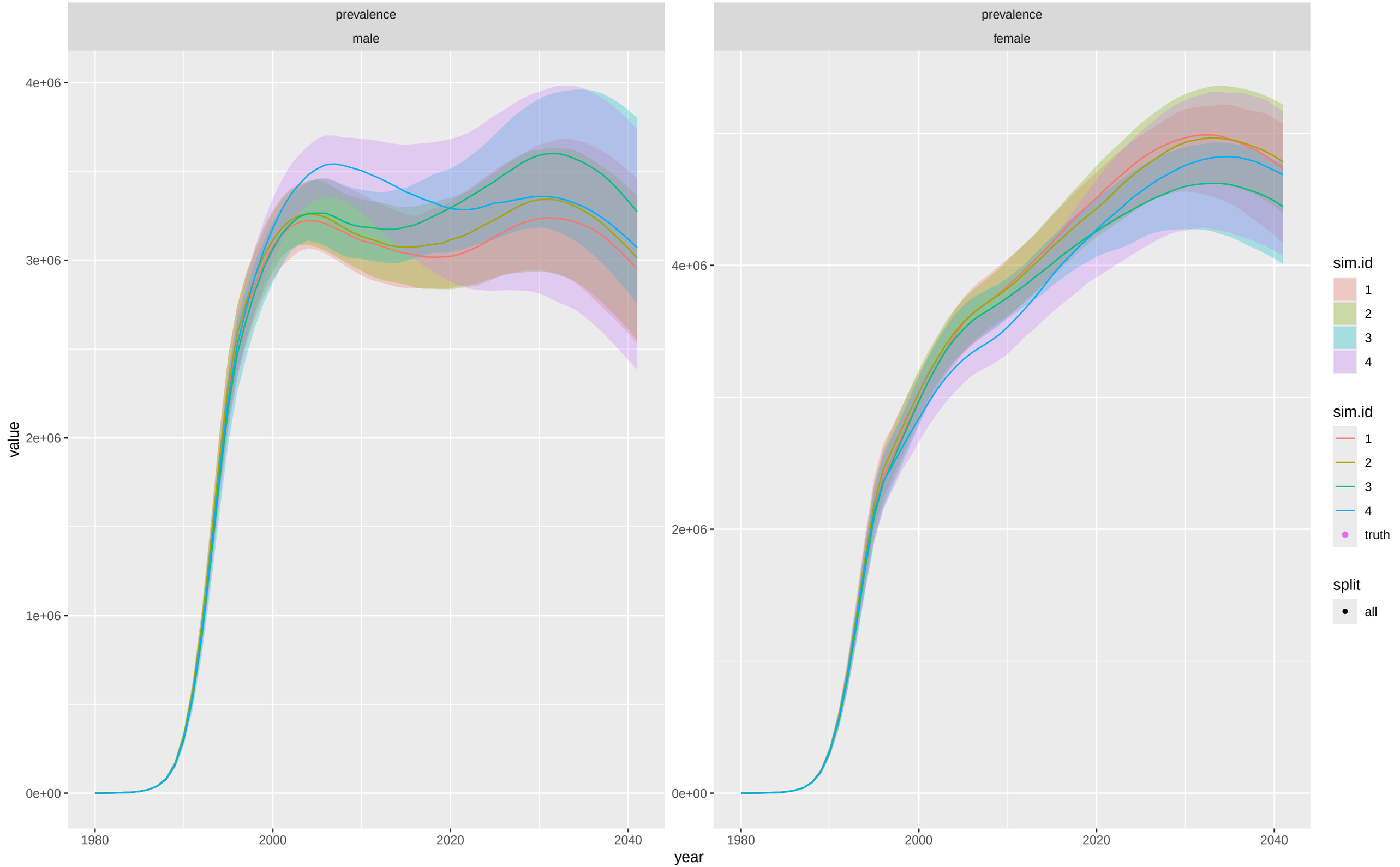
non.unaids.remainder step 2: cascade.by.age.sex (one color per chain)



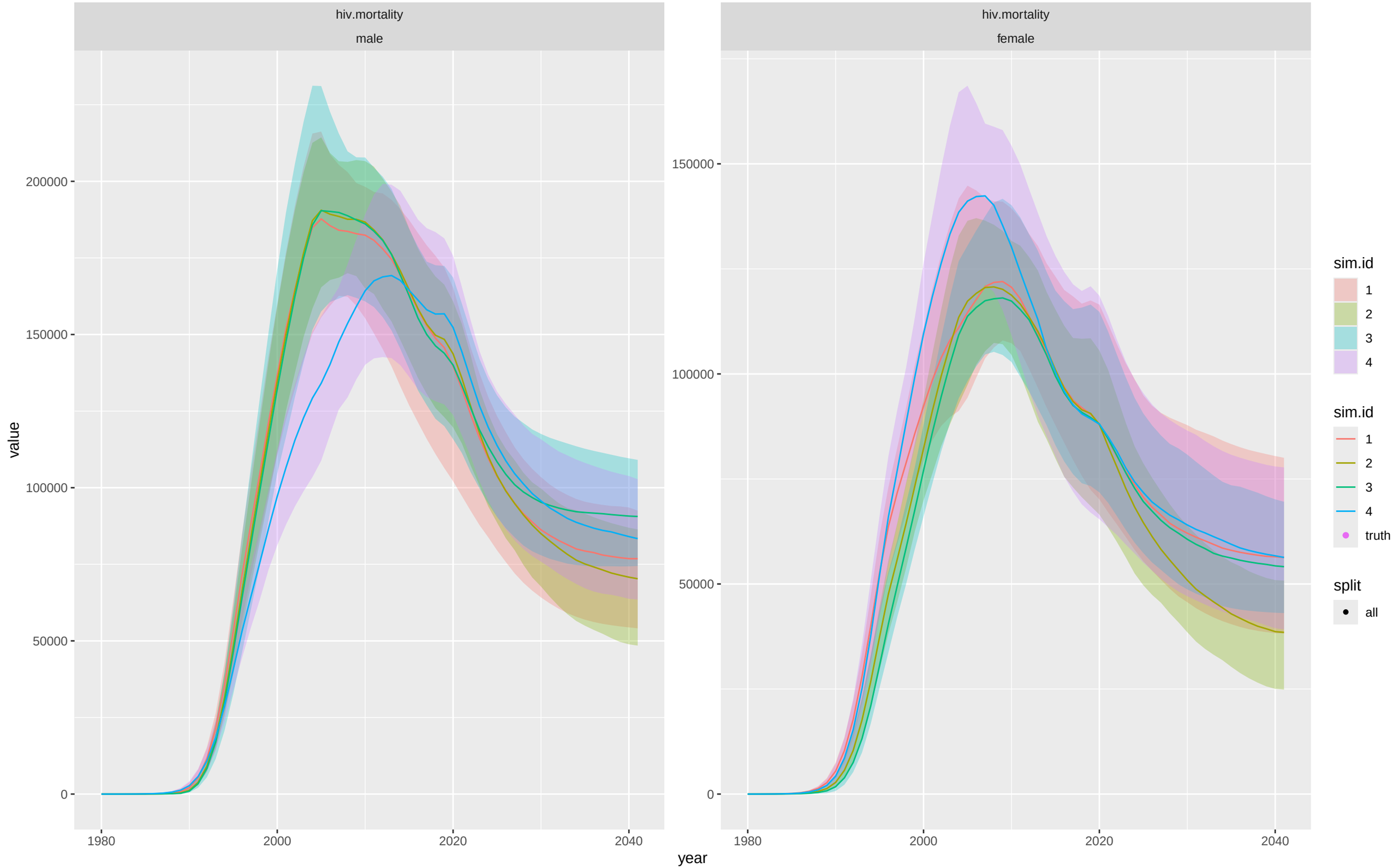
non.unaids.remainder step 2: incidence.by.sex (one color per chain)



non.unaids.remainder step 2: prevalence.by.sex (one color per chain)

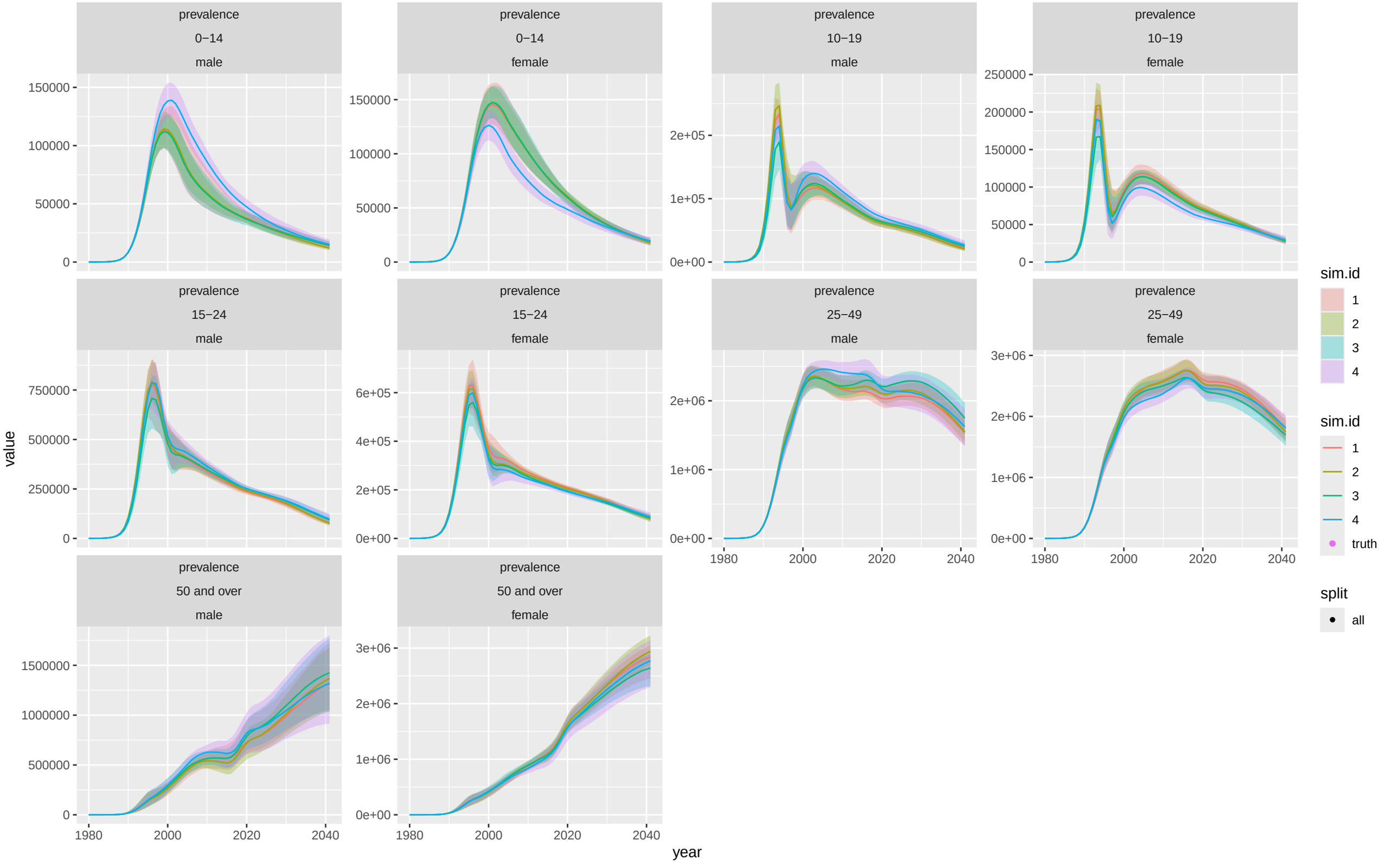


non.unaids.remainder step 2: hiv.mortality.by.sex (one color per chain)

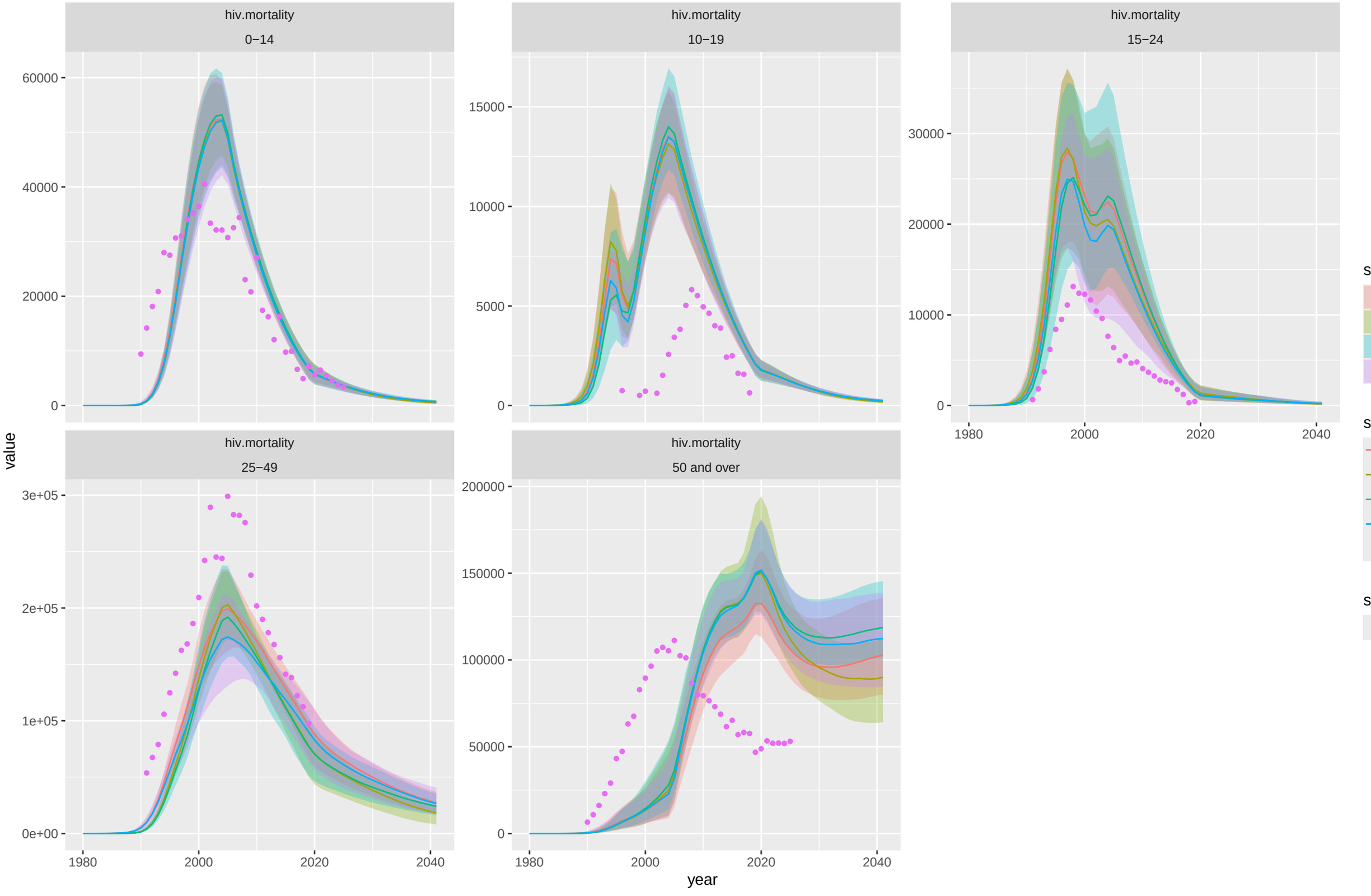


non.unaids.remainder step 2: incidence.by.age.sex (one color per chain)

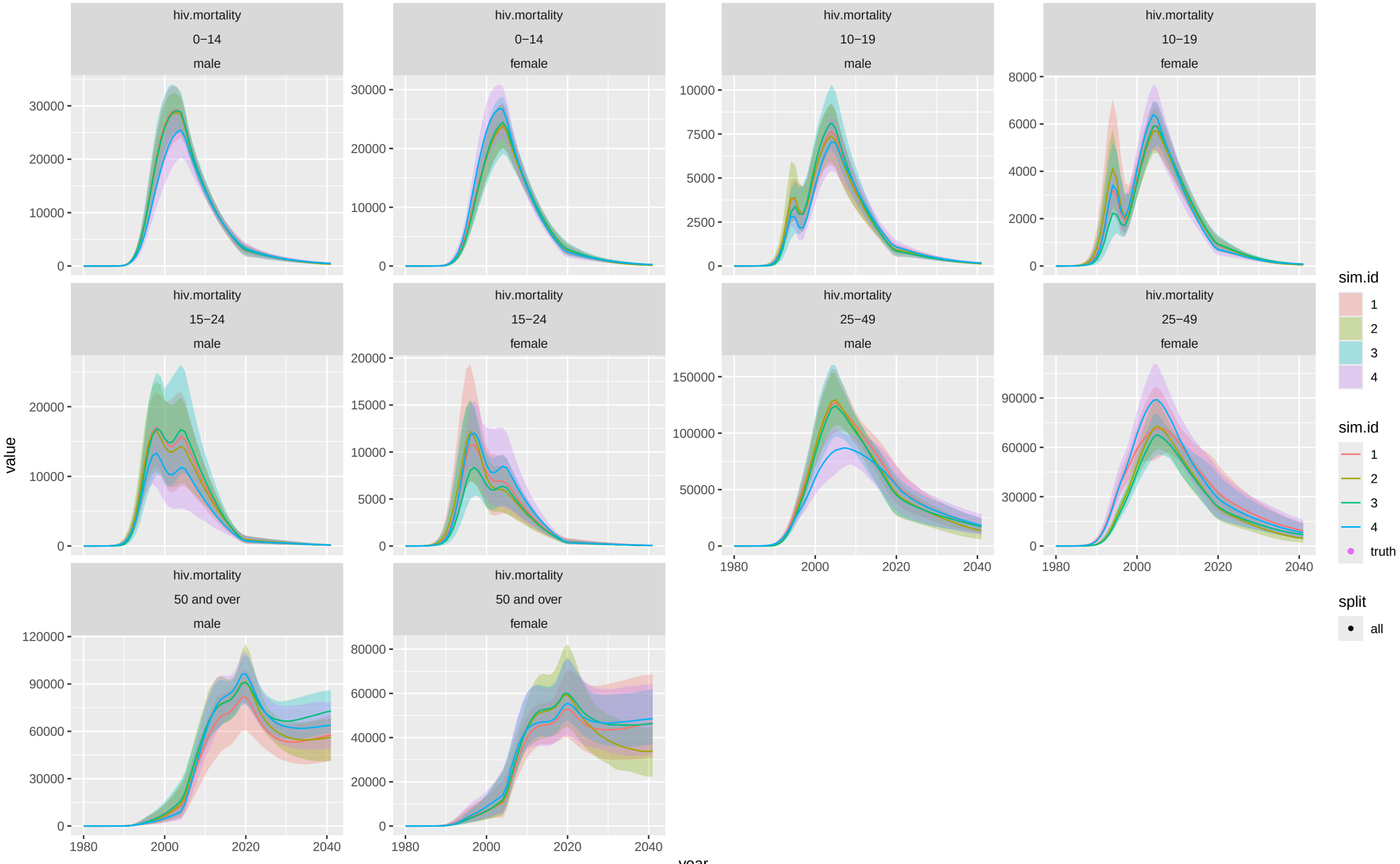
non.unaids.remainder step 2: prevalence.by.age.sex (one color per chain)



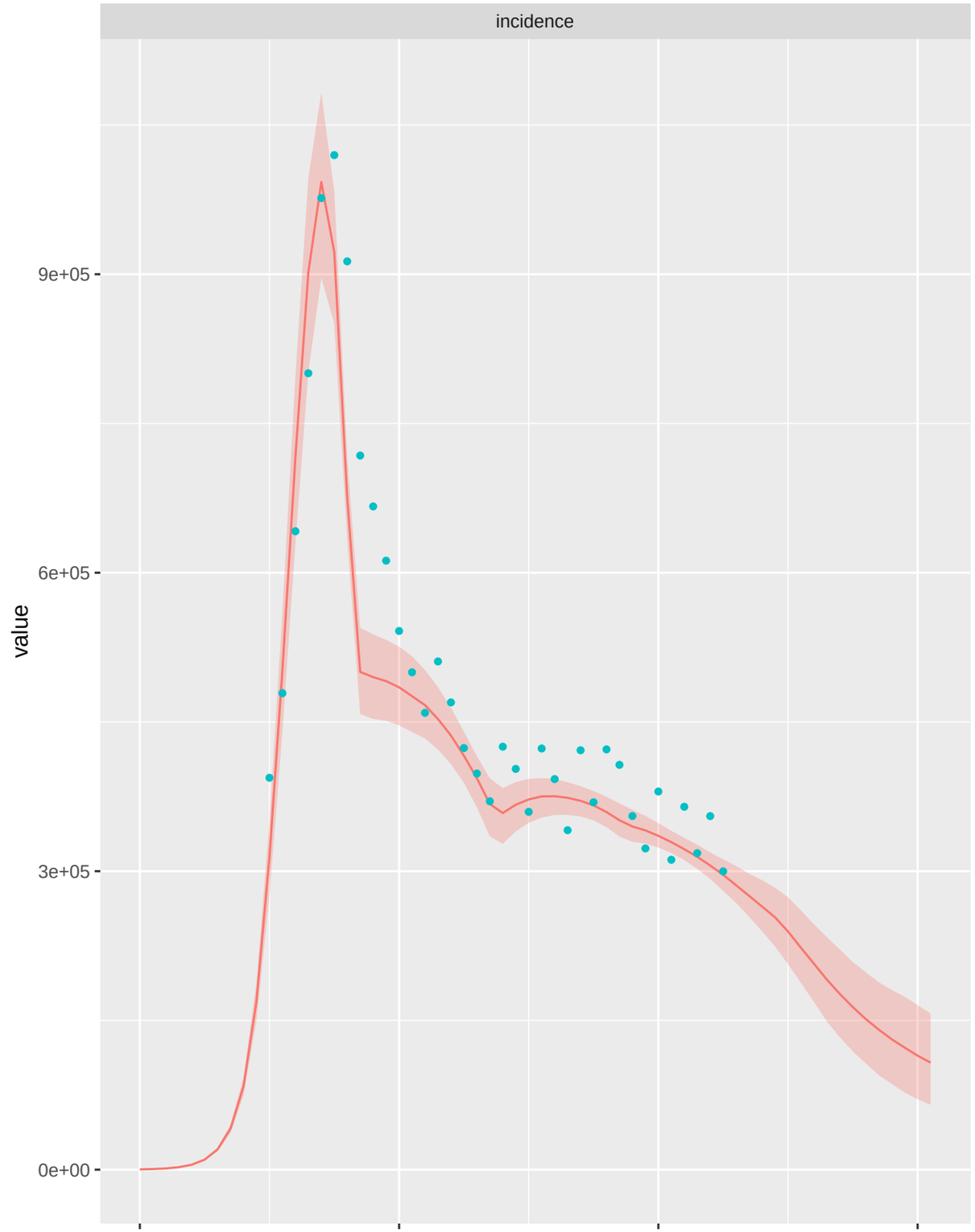
non.unaids.remainder step 2: hiv.mortality.by.age (one color per chain)

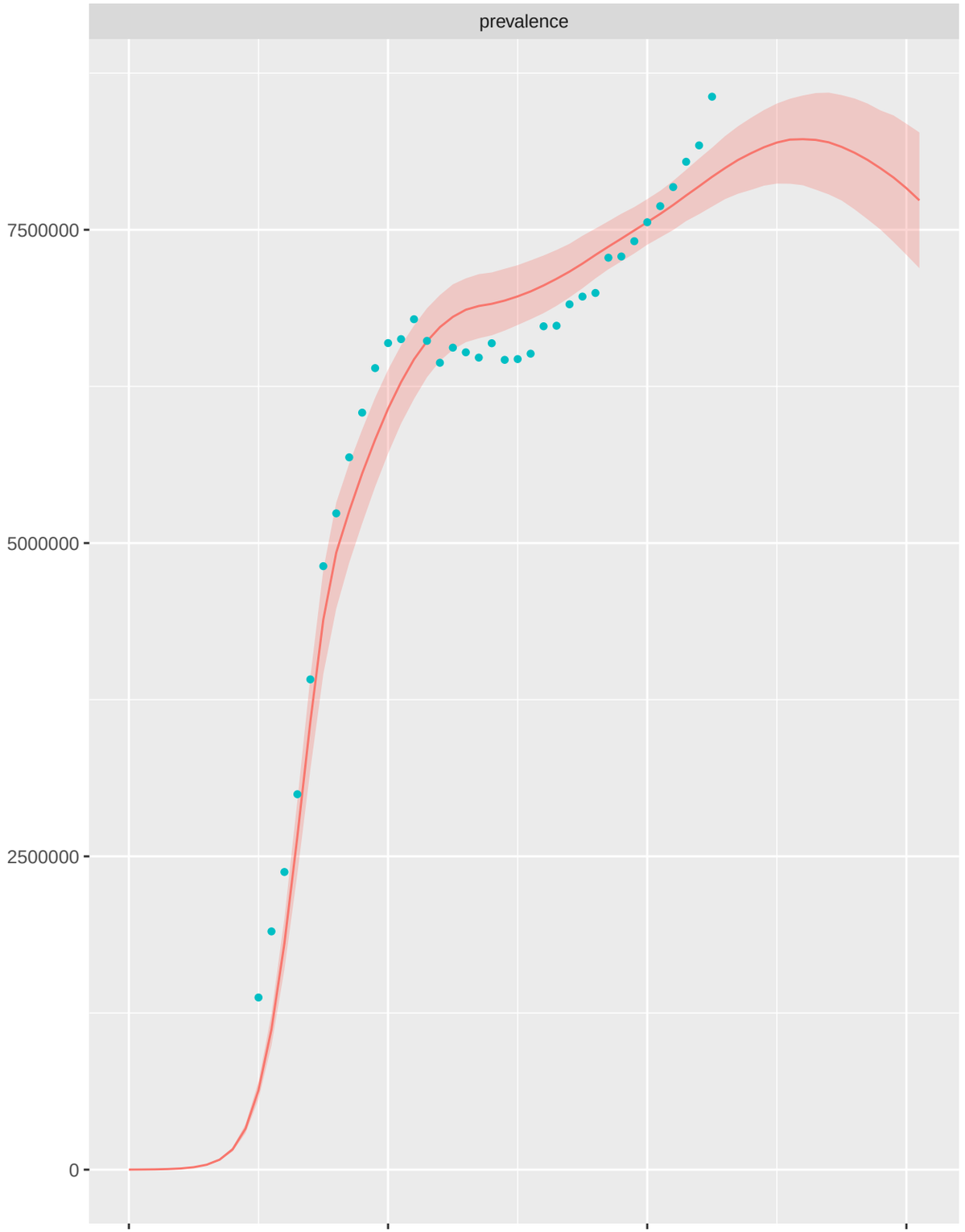


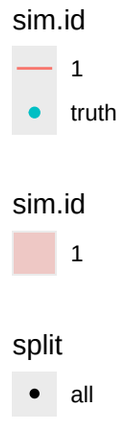
non.unaids.remainder step 2: hiv.mortality.by.age.sex (one color per chain)



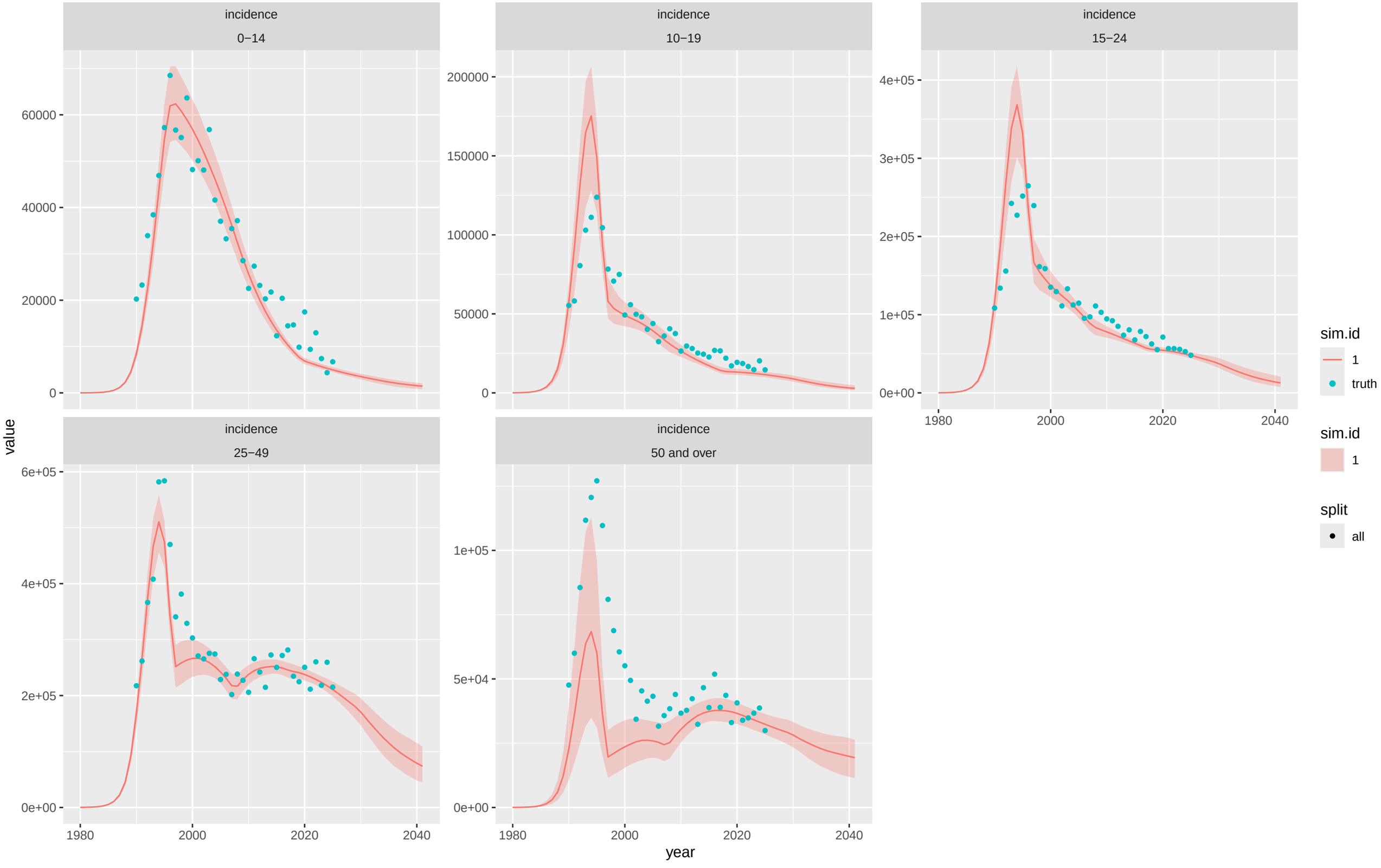
non.unaids.remainder: fit (merged Step 2 simset)



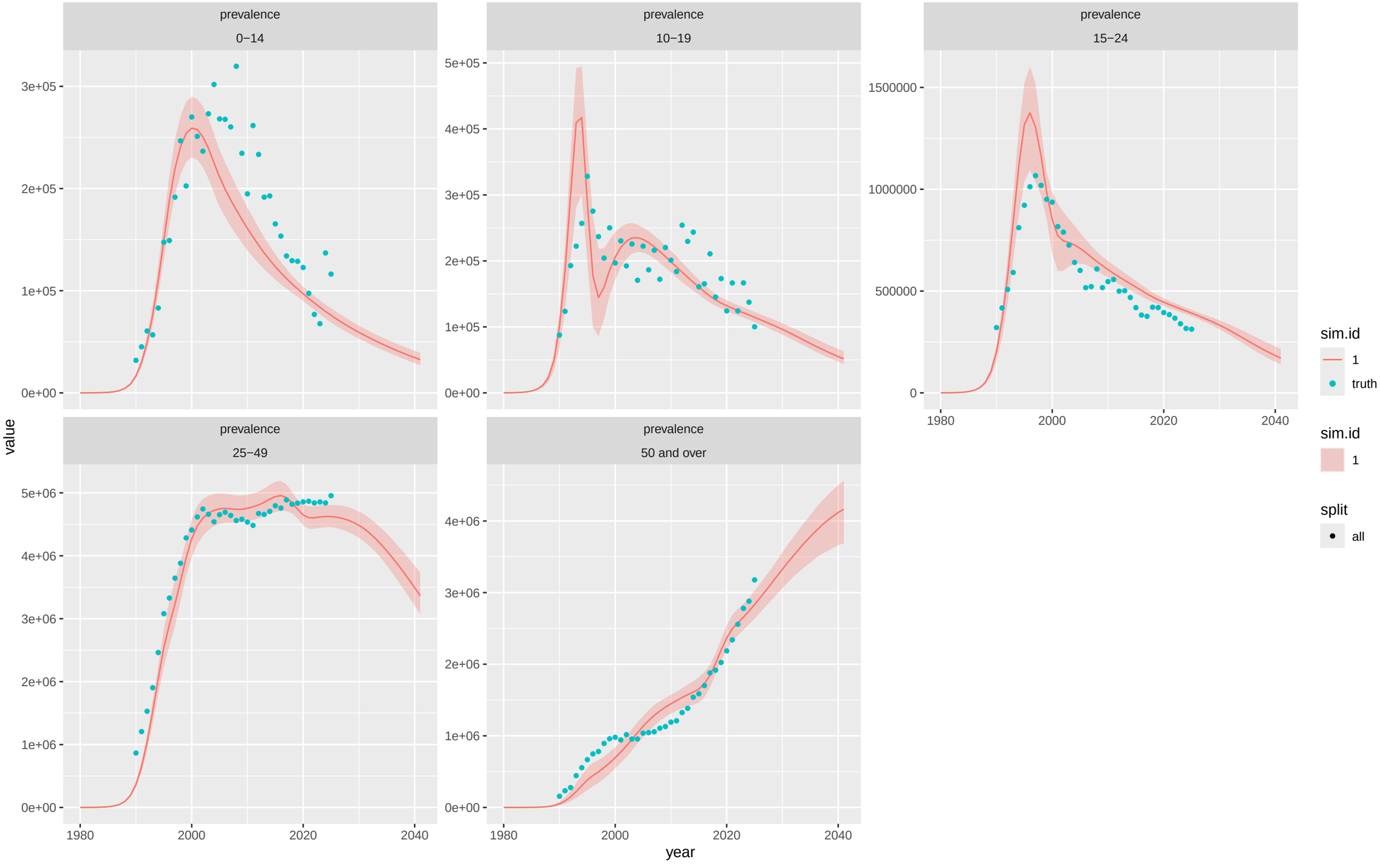




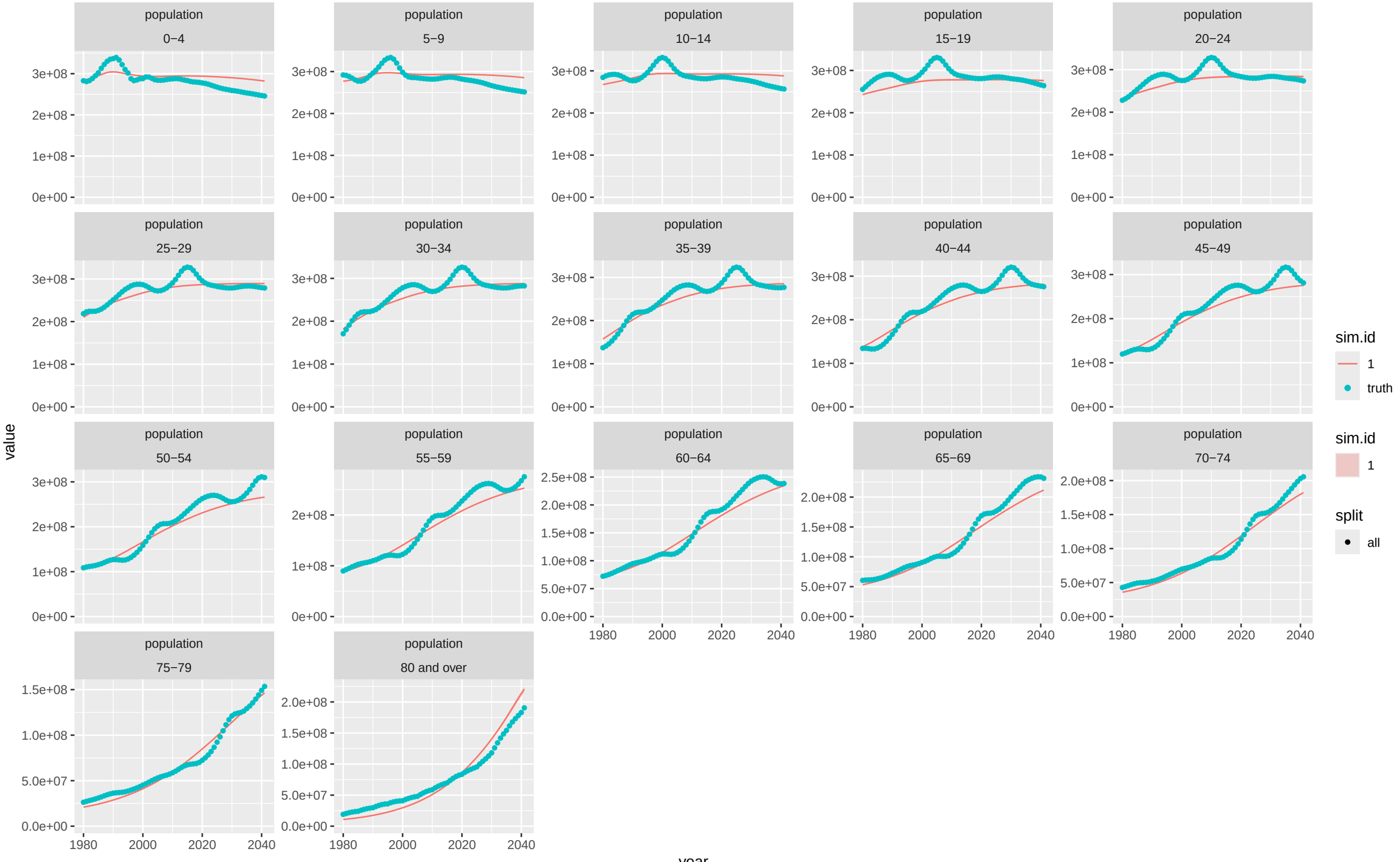
non.unaids.remainder: incidence.by.age (merged Step 2 simset)



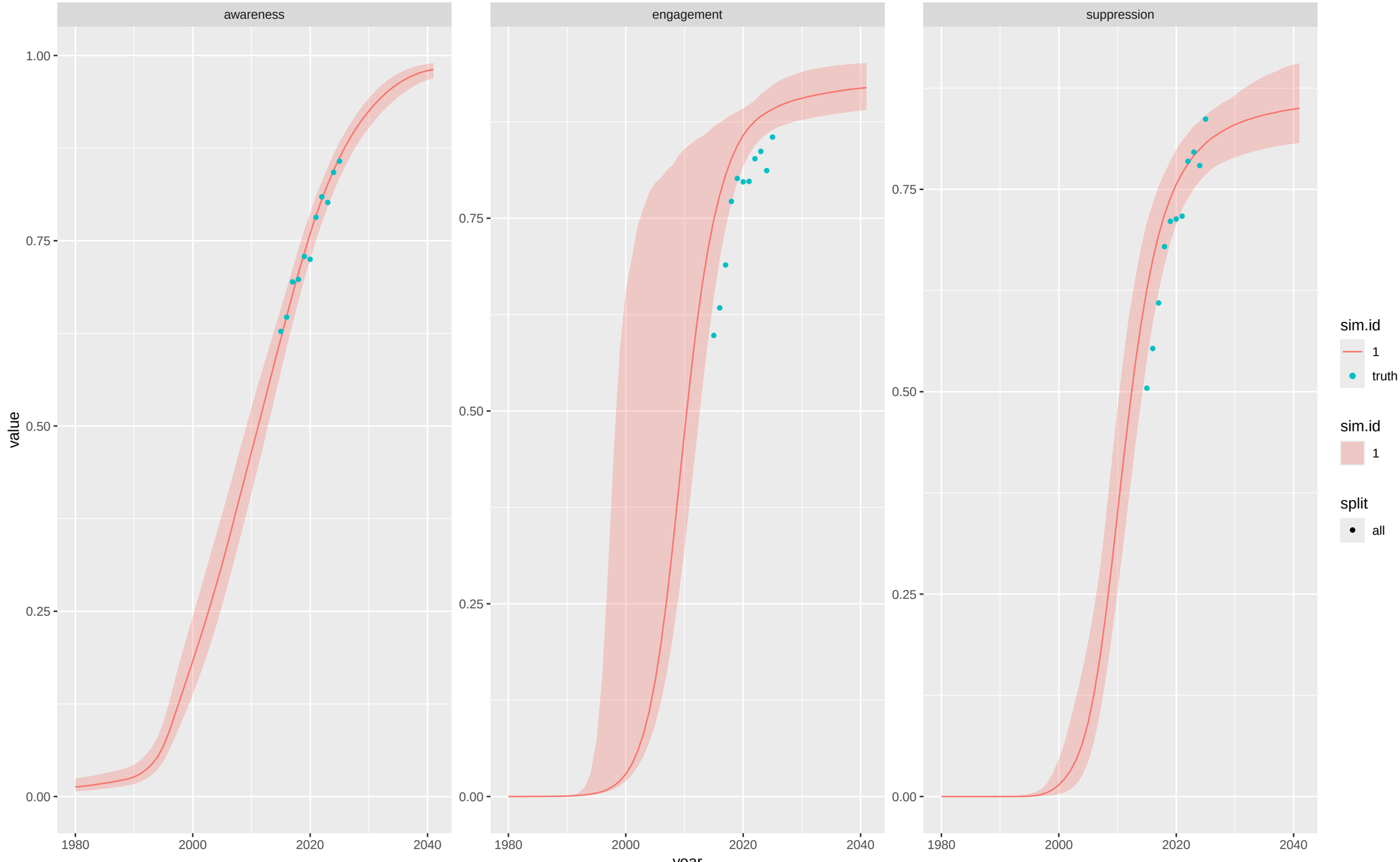
non.unaids.remainder: prevalence.by.age (merged Step 2 simset)



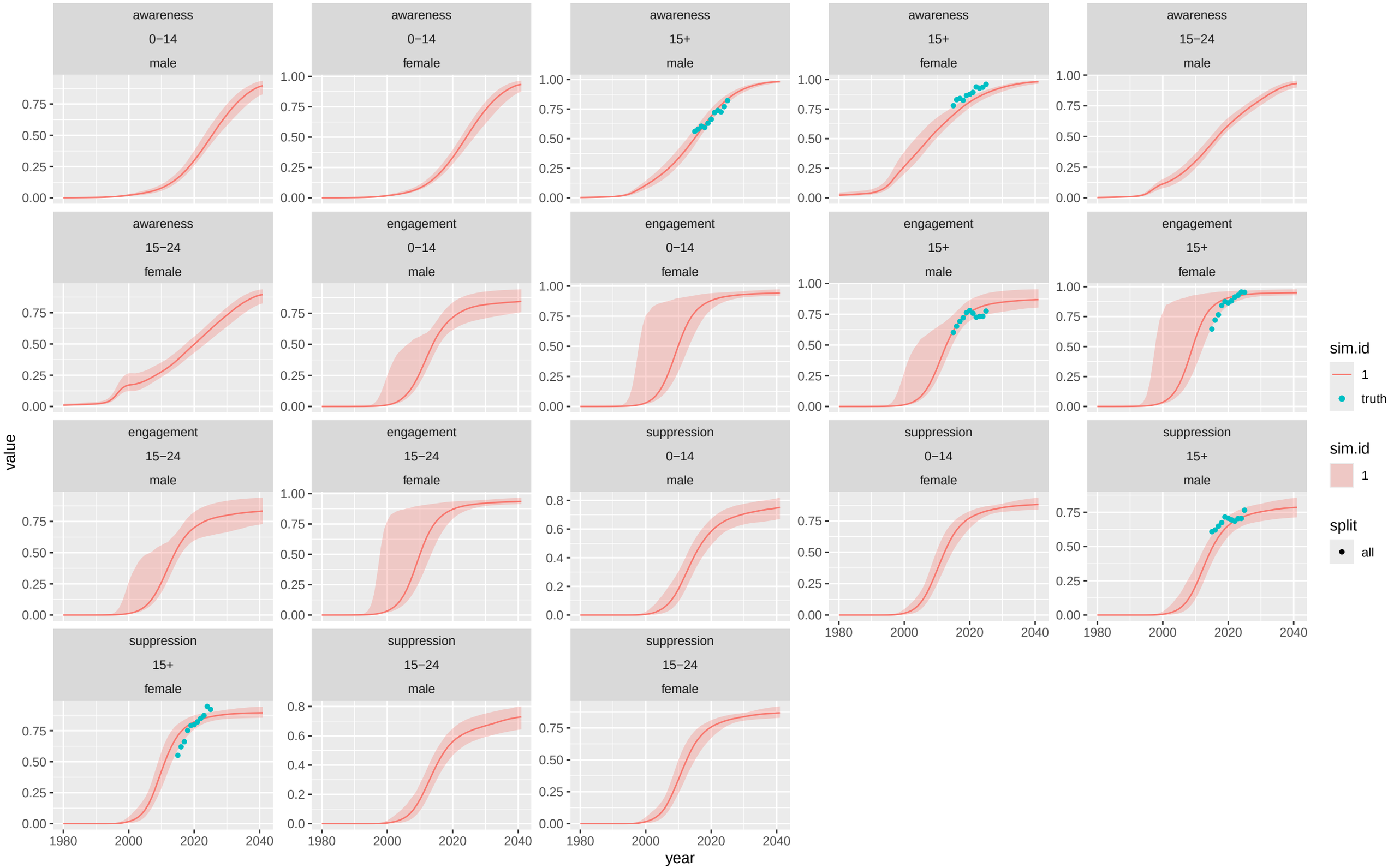
non.unaids.remainder: population.by.age (merged Step 2 simset)



non.unaids.remainder: cascade (merged Step 2 simset)



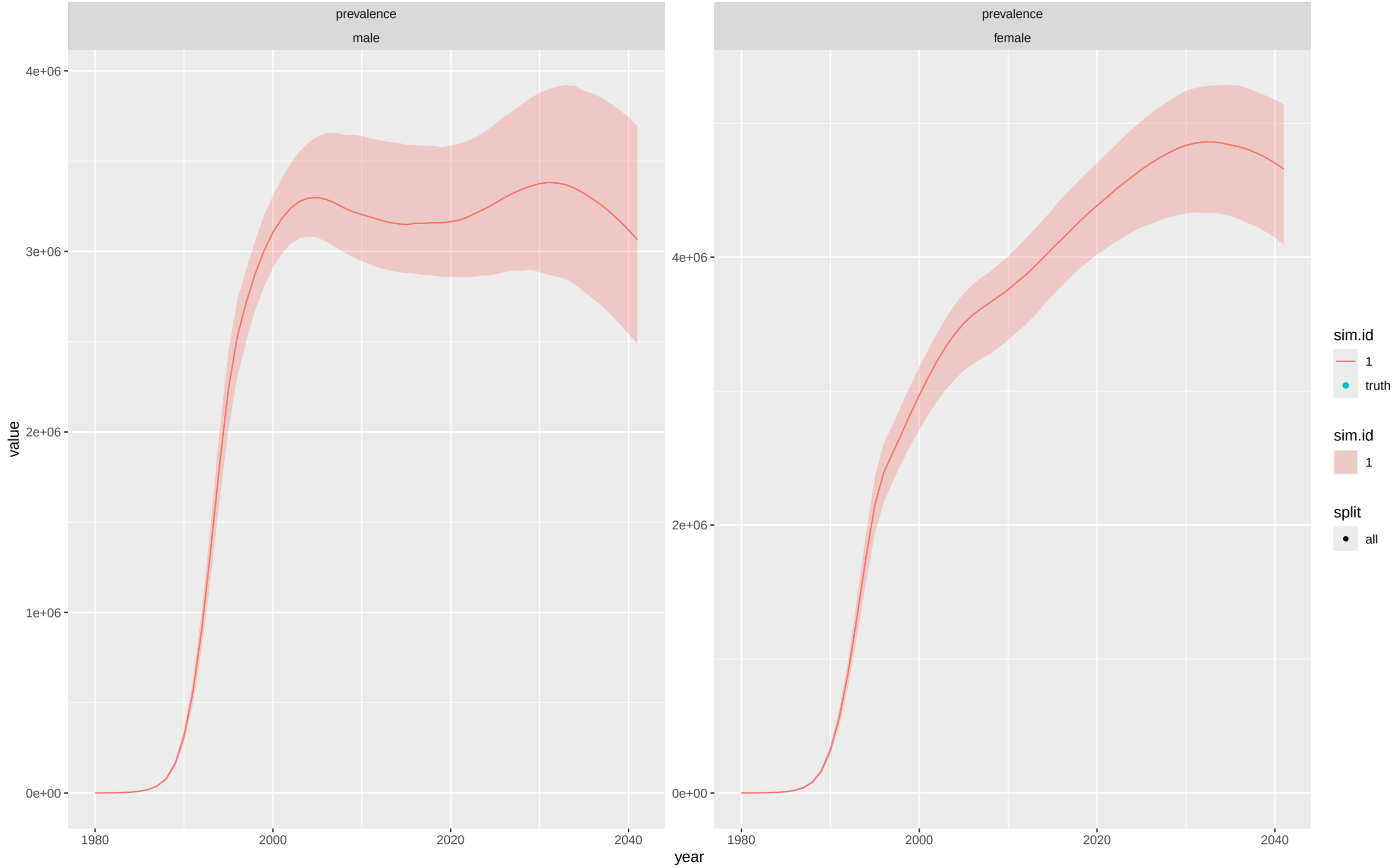
non.unaids.remainder: cascade.by.age.sex (merged Step 2 simset)



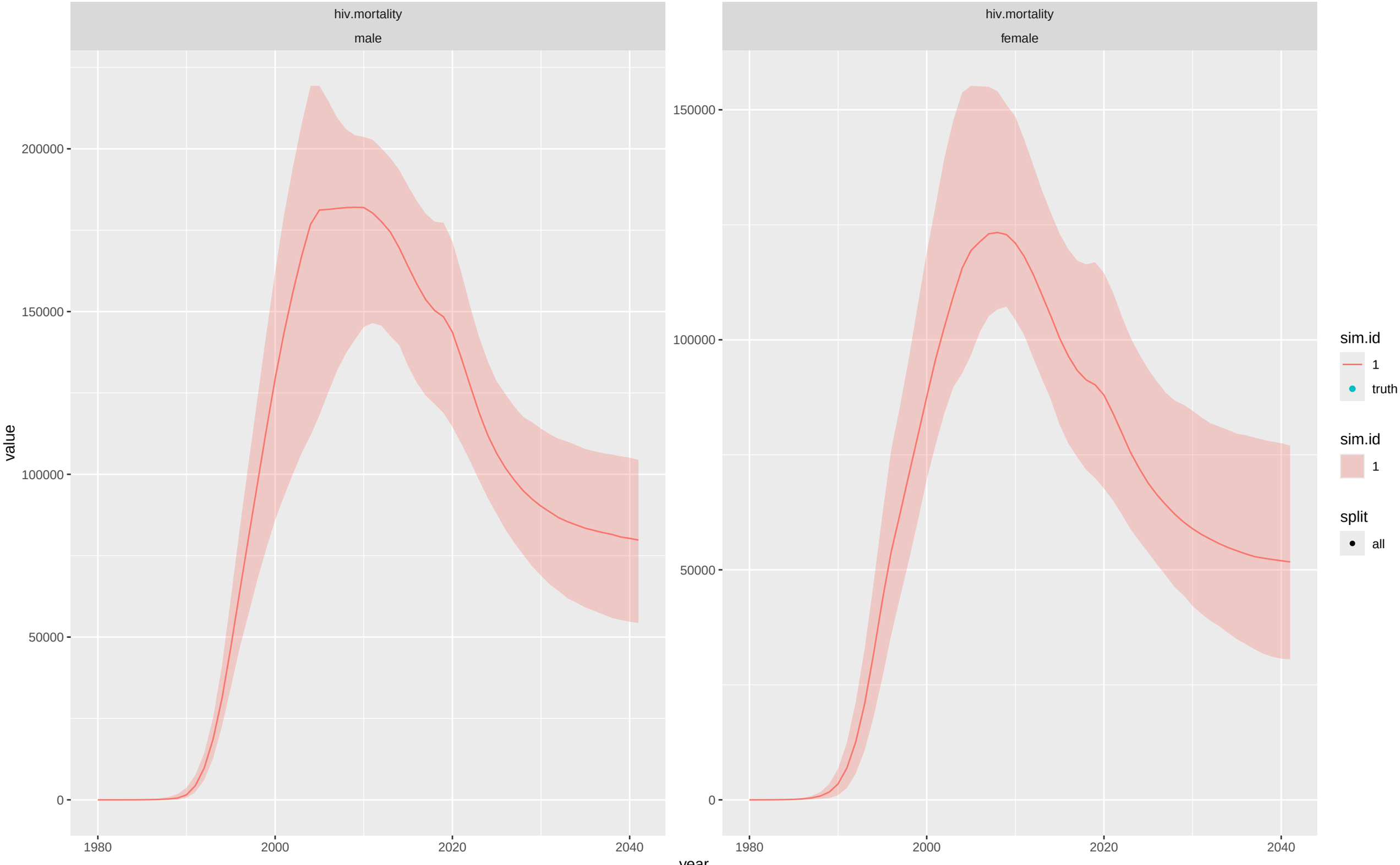
non.unaids.remainder: incidence.by.sex (merged Step 2 simset)



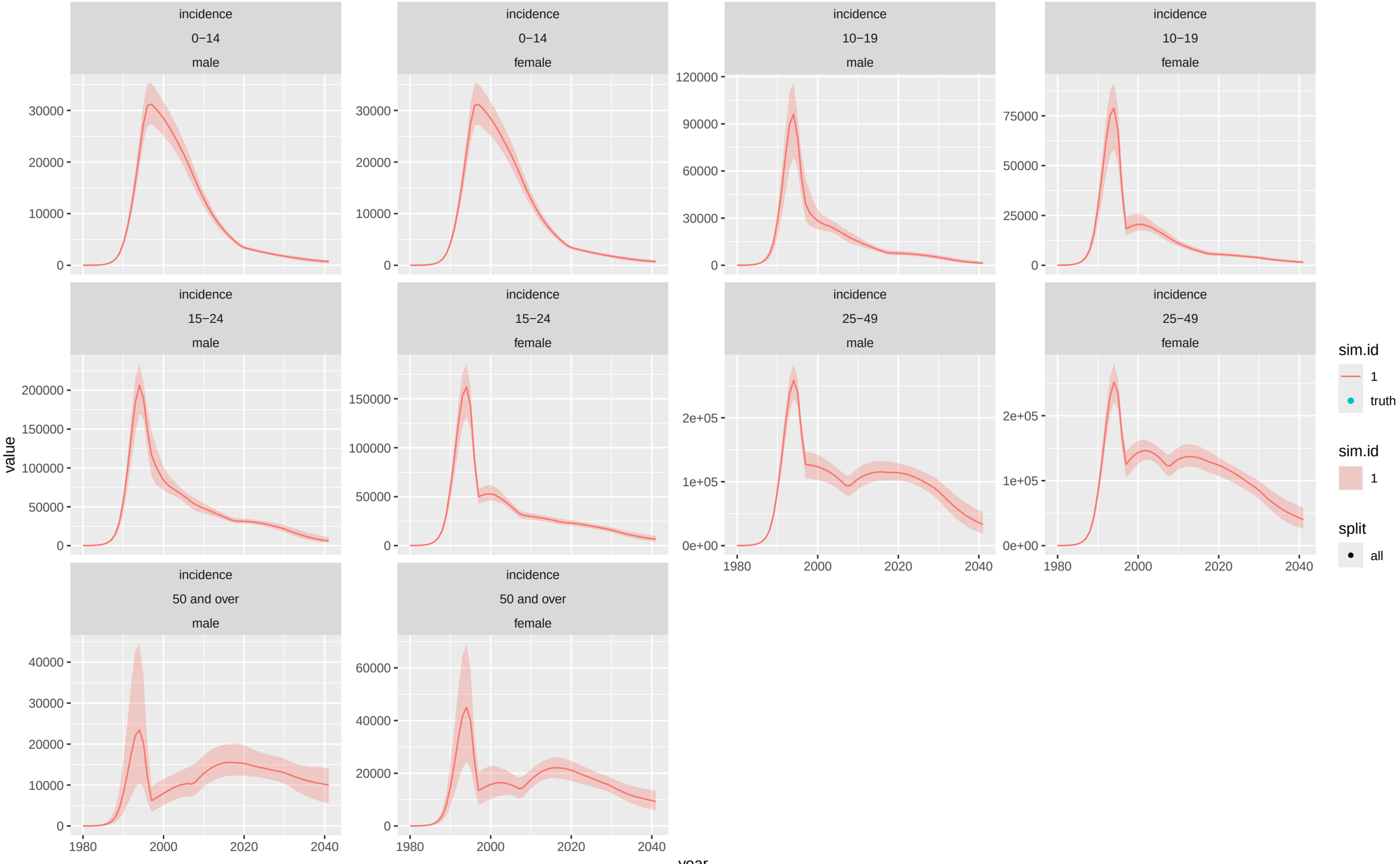
non.unaids.remainder: prevalence.by.sex (merged Step 2 simset)



non.unaids.remainder: hiv.mortality.by.sex (merged Step 2 simset)



non.unaids.remainder: incidence.by.age.sex (merged Step 2 simset)



non.unaids.remainder: prevalence.by.age.sex (merged Step 2 simset)

prevalence 0−14 male
prevalence 0−14 female
prevalence 10−19 male
prevalence 10−19 female
prevalence 15−24 male
prevalence 15−24 female
prevalence 25−49 male
prevalence 25−49 female
prevalence 50 and over male
prevalence 50 and over female

value

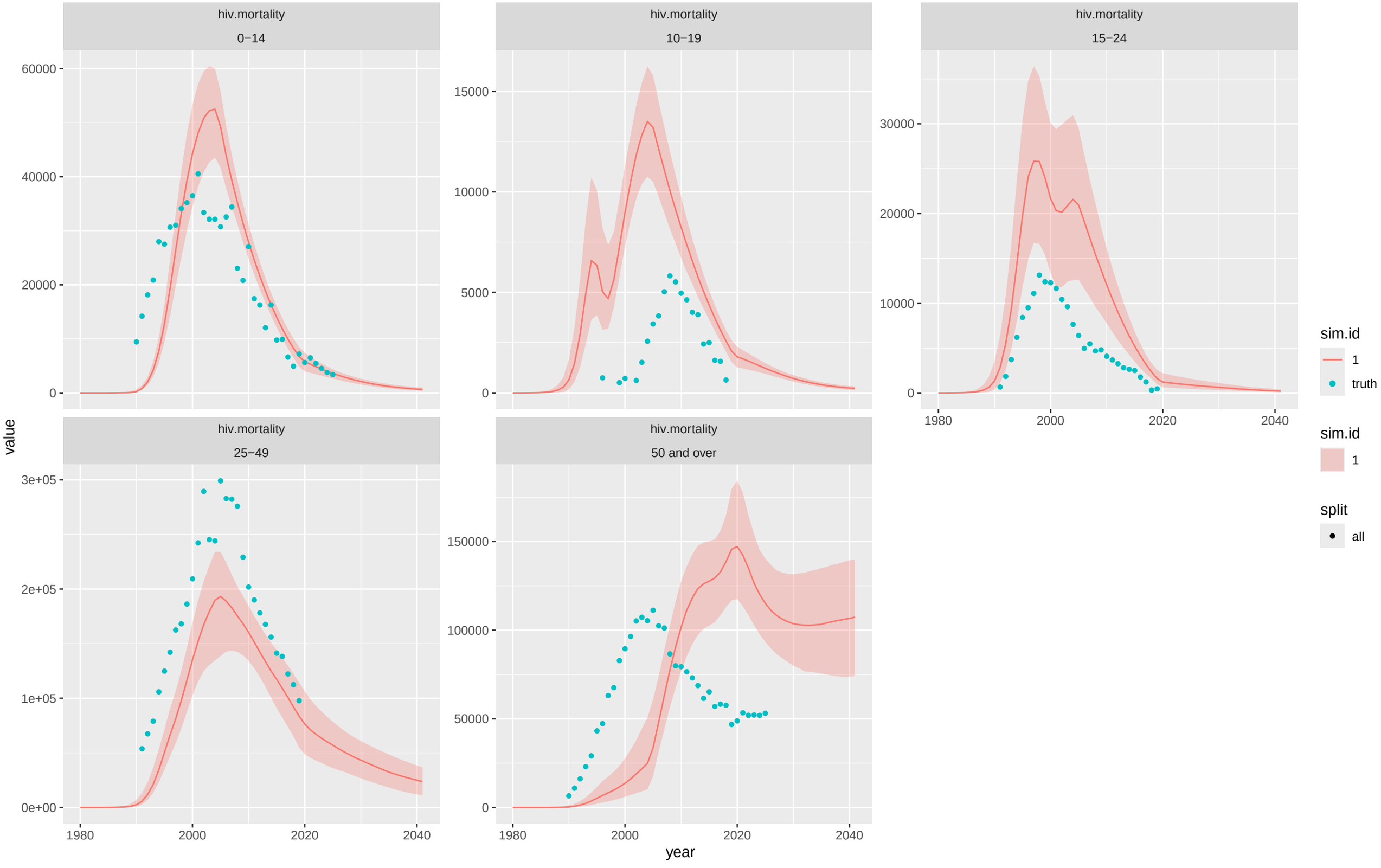
year

sim.id
1
truth

sim.id
1

split
all

non.unaids.remainder: hiv.mortality.by.age (merged Step 2 simset)



non.unaids.remainder: hiv.mortality.by.age.sex (merged Step 2 simset)

